Arista Access Node HSR-ACC-27: Optical Transport Failure Resolution SOP (ALARM_8385)

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_8385

Description

Optical Transport Failures (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Arista Access Nodes, specifically HSR-ACC-27, at HSR Layout Access Site, occur due to the inherent susceptibility of optical interfaces to environmental stressors, signal degradation, and physical damage. As part of the Arista's EOS (Extensible Operating System) architecture, the Access Node's control plane, responsible for forwarding and routing decisions, relies on the data plane's ability to transmit and receive optical signals through the SFP/XFP interfaces. When these interfaces malfunction or degrade, the control plane is affected, resulting in network performance issues and triggering the ALARM_8385 alarm. This SOP outlines the steps to diagnose and resolve such failures on the specified node, adhering to the Arista CLI dialect.

Pre-requisite Checks

- Pre-Requisite: Ensure remote hands are dispatched to HSR Layout Access Site to facilitate physical access to the node, if required., Description: Verify the presence of redundant links and confirm their operational status to avoid impacting network connectivity during the troubleshooting process.

- Pre-Requisite: Check for any recent software or configuration changes that may have contributed to the failure., Description: Verify the node's software version and ensure it is up-to-date, as newer versions may address known issues or vulnerabilities.

Potential Root Causes

Cause	Probability
SFP/XFP Interface Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss or Imbalance	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces gigabitEthernet 1/1 detail, Expected Output: Description: GigabitEthernet1/1, Port-Name: GE1/1, Serial-Number: ABC123, Part-Number: ABC123, HW-SW: 1.2.3, HW-Serial: 12345678, Description: This command displays detailed information about the specified GigabitEthernet interface, including its serial number, part number, and hardware and software versions.

Command: show system internal optics, Expected Output: Optics: 1, RX Power: -20.5 dBm, TX Power: 4.2 dBm, Description: This command provides real-time information about the node's optical interfaces, including the received and transmitted power levels, which can help
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identify potential issues with the SFP/XFP interfaces.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP Interface, Command: replace interfaces gigabitEthernet 1/1, Description: Use the 'replace' command to remove and replace the faulty SFP/XFP interface with a new one. Ensure the new interface is properly seated and secured before re-enabling it.

Step 2: Step: Adjust the Rx/Tx Power Levels, Command: set system internal optics rx-power -20.5 dBm tx-power 4.2 dBm, Description: Use the 'set' command to adjust the received and transmitted power levels on the affected interface. Consult the Arista documentation for the recommended power levels for the specific SFP/XFP interface being used.

Post-Check Verification

```
Command: show interfaces gigabitEthernet 1/1 detail, Expected Output:
Description: GigabitEthernet1/1, Port-Name: GE1/1, Serial-Number:
ABC123, Part-Number: ABC123, HW-SW: 1.2.3, HW-Serial: 12345678,
Description: Verify that the replaced or adjusted interface is
functioning correctly by checking its detailed information.

Command: show system internal optics, Expected Output: Optics: 1, RX
Power: -20.5 dBm, TX Power: 4.2 dBm, Description: Verify that the
adjusted power levels are within the recommended range for the
specific SFP/XFP interface being used.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: configure replace startup-config, Description: Use the 'configure replace' command to restore the node's original configuration, which will reset the affected interface to its previous state.

- Step: Revert to the Previous Software Version, Command: software install auto-revert, Description: Use the 'software install auto-revert' command to revert the node to the previous software version, which may have addressed the issue causing the failure.